Differential Equations Course Project

Grading Rubric

Base Project Score: 100 points

Written Report: 70 points   Video Presentation: 30 points.

The course project counts for 20% of the final course grade. An optional research poster may earn **10 bonus points** on the project score.

General Performance Levels

Level	General Description
Excellent	Mathematics is correct and clearly explained; modeling choices are justified; analysis and simulations directly address the research question; interpretation demonstrates strong understanding.
Proficient	Work is mostly correct and complete, with minor mathematical, computational, or explanatory gaps that do not substantially affect the main conclusions.
Developing	The main idea is present, but important mathematical steps, model explanations, simulations, or interpretations are incomplete, unclear, or contain notable errors.
Insufficient	Major required components are missing, the model is not adequately explained, serious mathematical errors prevent meaningful conclusions, or the work does not demonstrate independent understanding.

Part I: Written Research Report — 70 Points

Criterion	Full-Credit Expectations	Points
Research Question and Motivation	Clearly introduces the application, explains why it is worth studying, and states a focused research question that can be investigated with differential equations.	8
Background and Sources	Provides sufficient background for the reader; uses credible references; includes at least one scholarly source; accurately distinguishes cited information from the student's own work.	7
Model Formulation	Clearly defines all state variables, parameters, units, assumptions, and initial conditions. The differential equation or system is stated correctly. Every important term in the model is explained and connected to a modeling assumption or mechanism.	15
Mathematical Analysis	Uses appropriate course-level mathematics to analyze the model. Depending on the project, this may include analytical solutions, equilibria, phase lines, stability, long-term behavior, thresholds, conservation laws, or another appropriate analysis. Important steps are shown and mathematically correct.	15
Numerical Simulations and Computational Results	Includes meaningful numerical simulations that are mathematically consistent with the model. Figures have labels and captions. At least two useful simulations or comparisons are included. Parameter changes or scenarios are chosen for a clear reason rather than arbitrarily.	12
Interpretation, Discussion, and Limitations	Explains what the mathematical and numerical results mean in the original application. Connects conclusions back to the research question. Discusses assumptions, limitations, possible weaknesses, and reasonable future improvements.	8
Writing, Organization, and Citations	Report follows a clear research-paper structure, is readable and professional, uses mathematical notation appropriately, labels figures/equations when useful, and cites sources consistently. The student demonstrates ownership and understanding of the work.	5
Written Report Total		**70**

Part II: Recorded Video Presentation — 30 Points

The recorded presentation should be 10–15 minutes.

Criterion	Full-Credit Expectations	Points
Organization and Motivation	Introduces the topic efficiently, explains the motivation, and clearly states the research question. The presentation has a logical beginning, middle, and conclusion.	5
Explanation of the Model	Clearly explains the state variables, parameters, assumptions, differential equations, and initial conditions. The student explains where the major terms in the equations come from instead of simply displaying formulas.	8
Mathematical Analysis and Main Results	Accurately presents the most important mathematical analysis and demonstrates understanding of the methods used. The presentation focuses on the reasoning and main results rather than showing every algebraic step.	7
Simulations and Interpretation	Uses figures or simulations effectively and explains what they show. Connects computational results to the model and the research question.	5
Communication, Visual Quality, and Timing	Speaks clearly and professionally, uses readable visual material, explains rather than reads the report verbatim, and remains within the 10–15 minute target.	5
Video Presentation Total		**30**

Base Project Score

$$\boxed{\text{Base Project Score} = \text{Written Report Score} + \text{Video Presentation Score.}}$$

Therefore,

$$\boxed{70 + 30 = 100 \text{ possible base points.}}$$

Optional Research Poster Bonus — +10 Points

Students may submit an additional research poster. A poster that meets the expectations below earns

$$\boxed{+10 \text{ bonus points.}}$$

The poster should:

- have a clear title and author name;
- state the research question and motivation;
- summarize the model variables, parameters, and differential equations;
- include the main mathematical result;
- include at least one meaningful simulation or figure;
- summarize the primary conclusions;
- identify important model limitations;
- include appropriate references;
- be visually organized and readable as a research poster.

The bonus is added to the base project score. For example,

$$88/100 + 10 = 98/100,$$

and

$$95/100 + 10 = 105/100.$$

The project remains weighted as 20% of the overall course grade.